

TRI-NETRA ADCS: design loop, SILS, Monte Carlo and soft OILS

From each customer case to a sized, selected and dispatched attitude-control configuration, verified on the Rust engine (POP v51 in the loop), against the MATLAB twin, and with the flight software running as Cortex-M4F firmware.

AIS_3U SELECTED mtq_fmr feasible	AIS_IMG_3U SELECTED mtq_fmr feasible	ENGINE VS MATLAB VERDICTS 94/107 truth environment identical
SILS VS SOFT OILS VERDICTS 105/107 worst OBC load 6.8 %	VIRTUAL OBC BIT-IDENTICAL 35 runs, process + QEMU	CAMPAIGNS ON THE ENGINE 8 MC + edge, vs MATLAB

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Repository revision	1ce136c
Engine	adcs-engine-rs/1.2.0 (adcs-case/1, POP v51 port in-loop, adcs-design, soft OILS)
Flight software	trinetra-fsw-c/1.0.0 and trinetra-fsw-rs (C99 and Rust no_std, adcs-fswcfg/1)
Cases	ais_3u, ais_img_3u
Regenerate	<code>python3 tools/pipeline.py · python3 tools/engine.py campaign · python3 tools/engine.py oils · python3 tools/vv_report.py</code>

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1. Summary and findings

ais_3u (coarse class): the loop selects `mtq_fmr` (Magnetorquers + fluid momentum loop), feasible, after 4 iteration(s). Methods: `detumble = mtq`, `sun_acquisition = fmr`, `sun_referencing = fmr+mtq`, `nadir_pointing = fmr+mtq`. Budget: 1.446 kg, 0.537 L. Best benchmark (catalogue actuators, same rule): `mtq_rw`, closest (not feasible), 0.393 kg — `sun_acquisition: power_mean (power)`; `sun_referencing: power_mean (power)`; `nadir_pointing: power_mean (power)`.

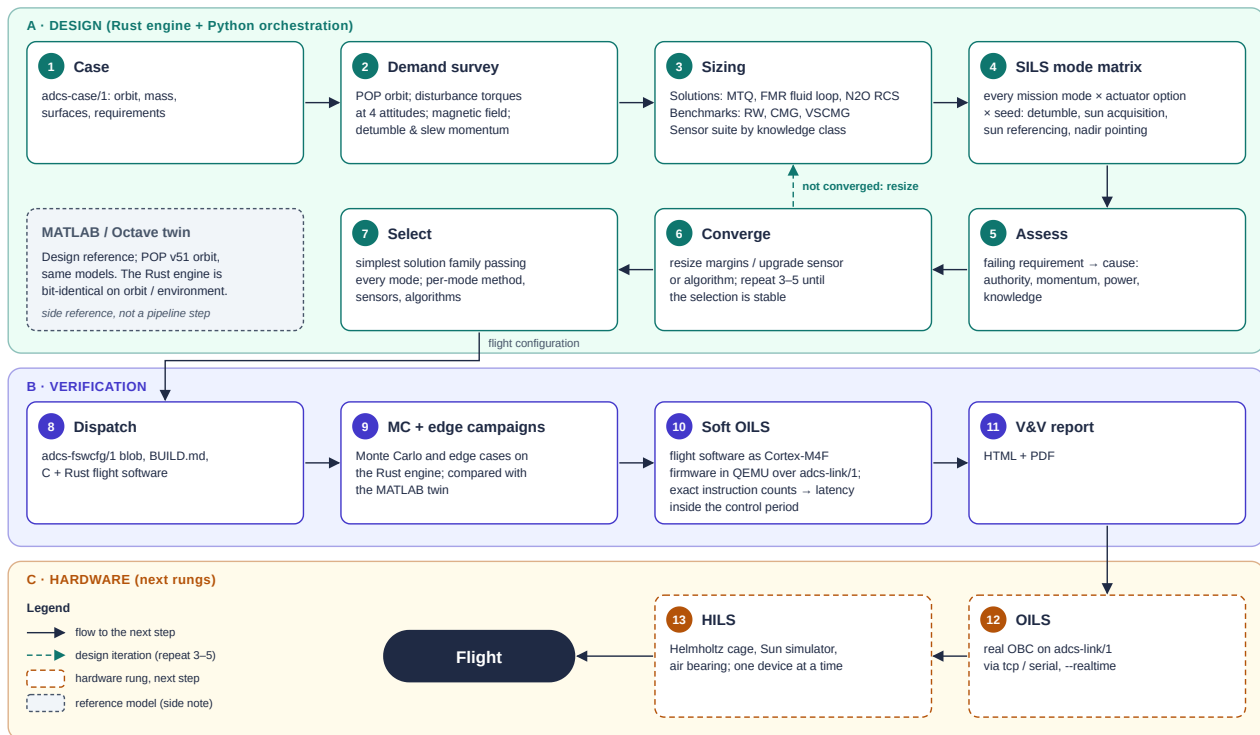
ais_img_3u (fine class): the loop selects `mtq_fmr` (Magnetorquers + fluid momentum loop), feasible, after 7 iteration(s). Methods: `detumble = mtq`, `sun_acquisition = fmr`, `sun_referencing = fmr+mtq`, `nadir_pointing = fmr+mtq`. Budget: 1.596 kg, 0.534 L. Best benchmark (catalogue actuators, same rule): `mtq_rw`, feasible, 0.996 kg.

2. Architecture and flow

The design loop sizes every actuator option to the case, flies every mission mode with every option in the SILS, converges the sizing, sensors and algorithms, selects the lightest of our solution families that passes, dispatches its flight configuration and verifies it by Monte Carlo and in soft OILS. The same flight software (C99 and Rust, one pseudocode contract) runs in every rung behind one byte-level HAL.

TRI-NETRA ADCS: from customer case to flight

End-to-end V&V flow: design loop, software verification, then hardware-in-the-loop rungs



TRI-NETRAADCS - Agastya

Figure 1. From customer case to flight: design loop, verification, hardware rungs.

Who runs what: languages and layers

TRI-NETRA ADCS, from the flight contract at the top to the ground tooling and outputs at the bottom

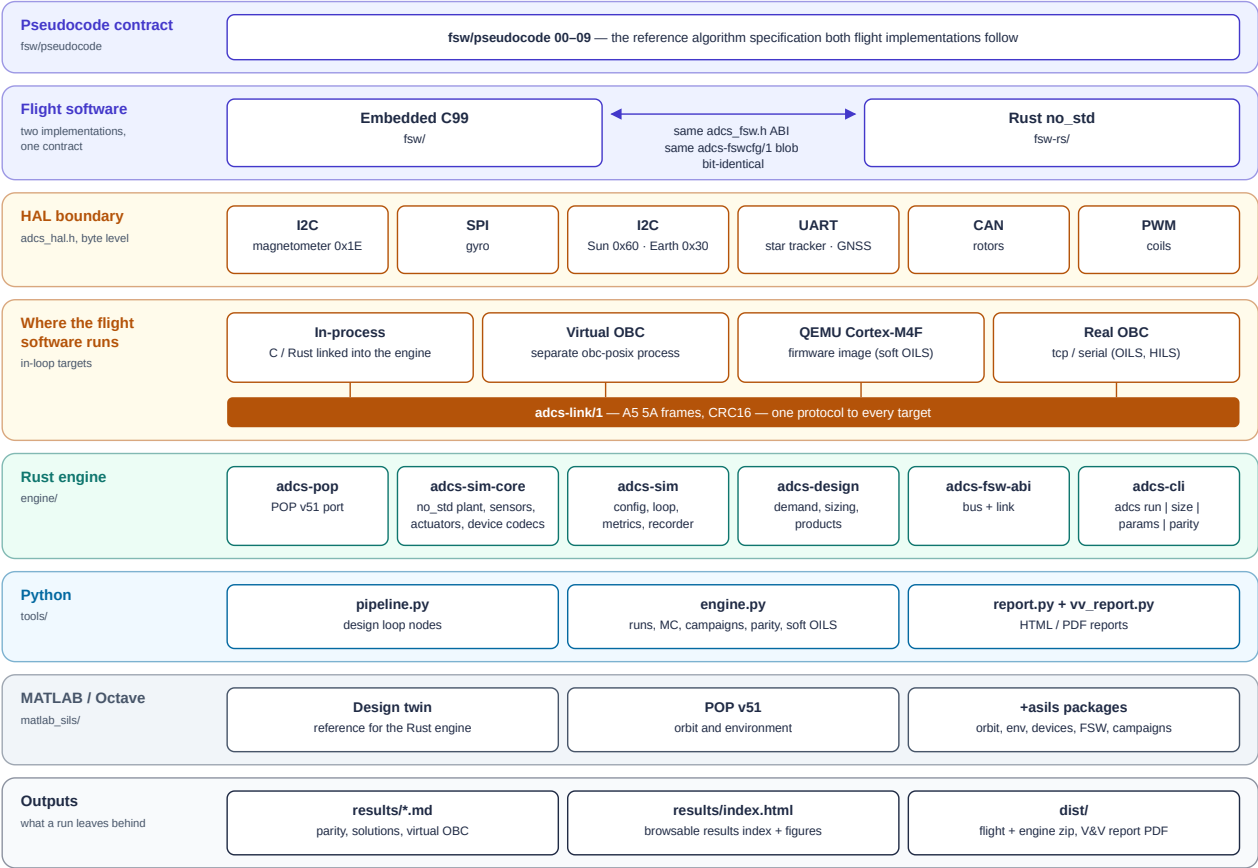


Figure 2. Languages and layers.

3. Nodes

Every step of the chain is a node with its inputs, outputs, parameters and rules in one registry (`matlab_sils/data/pipeline/nodes.json`; `docs/NODES.md`). The pipeline reads its parameters from it, so this table is what ran.

#	node	stage	runs in	what it does	parameters	rules
1	case	input	file	The customer case: orbit, mass properties, surfaces, requirements and budgets (adcs-case/1).	—	every requirement names its source (stated by the user, or UNCONFIRMED with a plan reference) the case file is part of every cached run's key: a changed requirement re-flies the matrix
2	catalogue	input	file + Python (tools/catalogue.py derives the modelling block)	Bought actuators as their datasheets state them: CubeSpace, AAC Clyde Space and Rocket Lab reaction wheels, Tensor Tech control moment gyroscopes. One file per model with the datasheet numbers, the source URL and the model parameters derived from them (each derived value says how).	vendors = CubeSpace, AAC Clyde Space, Rocket Lab, Tensor Tech	a number not on the datasheet is null in the datasheet block, never guessed model parameters the engine needs but the datasheet omits (rotor inertia, friction, imbalance) are derived and marked derived, with the rule RW, CMG and VSCMG are bought: they are selected from the catalogue, never sized by a law

#	node	stage	runs in	what it does	parameters	rules
3	demand	design	Rust (adcs-design::demand)	What the case asks of an actuator: one orbit on the POP orbit at four held attitudes with the SILS torque models; peak disturbance, cyclic and secular momentum, weakest field, detumble and slew momentum.	k_h = momentum margin: 2, or 1/(1 - req.hsat) k_tau = 1.5	h_req = k_h x max(cyclic + secular per orbit, slew momentum) tau_req = k_tau x max(peak disturbance, slew torque) fine class when req.ake <= 0.05 deg (star tracker, precision gyro)
4	size_mtq	design	Rust (adcs-design::mtq)	Our magnetorquer coils, sized: the largest of the dumping, momentum and detumble dipoles.	floor_Am2 = 0.05 anchor = SYN-CT-1: mass 0.03 kg, 0.3 W per 0.45 Am2	m = max(2 tau_dist/(0.5 B_min), 2 h_secular/(0.3 B_mean T), h_detumble/(0.3 B_mean t_detumble/2)) the coils-only family carries the pointing-grade coil (at least twice the dumping dipole)
5	size_fmr	design	Rust (adcs-design::fmr, empump)	Our fluid momentum loop, one ring per axis, designed with its electromagnetic DC conduction pump (no permanent magnet).	electrode_current_max_A = 10 iron_flux_T = 1.2 fluid = galinstan cruise_fraction = 0.4	grid over flow speed, bore, gap field and pump length; keep the least mass + lambda x steady power that meets h_req and tau_req torque capacity = I_max B / b x 2 S A / L; coil m_cu = sqrt(lambda K)
6	size_rcs	design	Rust (adcs-design::rcs)	Our N2O cold-gas RCS, 6 couples: thrust class and propellant tank.	isp_budget_s = 60 thrust_classes_N = 0.005, 0.01, 0.02, 0.05, 0.1 meop_bar = 70	thrust: the smallest class above the couple the detumble and slew need propellant: detumble + dumping over life, 1.2 margin

#	node	stage	runs in	what it does	parameters	rules
7	<code>select_rotor</code>	design	Rust (adcs-design::catalogue)	The benchmarks' momentum actuators chosen from the catalogue: for RW three orthogonal wheels, for CMG and VSCMG a four-unit 54.74 deg pyramid.	rw_units = 3 cmg_units = 4 pyramid_deg = 54.74 per_unit_need = rw: h_max >= h_req x scale and torque >= tau_req x scale (three wheels, one per axis); cmg: h_max >= h_req/2 x scale and torque >= tau_req/2 x scale (a pyramid puts about two units on any axis); vscmg: as cmg; the rotor runs at half its momentum so its speed can move both ways choose = the lightest model that meets the need; ties by steady power, then volume selectable = a catalogue model whose datasheet states momentum, torque, mass and steady power (tools/catalogue.py)	when no model meets the need the largest is taken and the gap is reported (the loop cannot size it up) an authority increase from the converge node raises the need, so the next model up is chosen
8	<code>size_sensors</code>	design	Rust (adcs-design::size_all)	The sensor set of every product: magnetometer, six sun sensors, gyro, GNSS, Earth sensor; the star tracker on the fine class or when knowledge asks for it.	star_tracker = SYN-ST-1 st_heads_default = 2 gyro_fine = TRN-GYRO-P1 gyro_coarse = SYN-GYRO-1	gyro grade g: noise x g, mass and power / g (fibre-optic class below 1)
9	<code>budget</code>	design	Rust (adcs-design::budget)	Each family's product: mass, steady power and volume summed over every fitted unit.	—	units per fill = number of axes / spin axes / boresights; the six sun sensors count as one part

#	node	stage	runs in	what it does	parameters	rules
10	matrix	SILS	Rust engine + C flight software (adcs run)	Every mission mode x actuator option x seed on the sized products; options that fail on performance also fly every algorithm of their slot.	seeds = 1, 2 cache_key = scenario, product, parts, seed, flight-software build, case file algorithm_candidates = detumble: bdot_gyro, bdot_mag, bdot_bangbang, genbdot_l1; sun_acquisition: sunspin_damped, sunspin_l1l2, sunspin_l1l2_e2; mtq_pointing: mtq_pd, mtq_lqr, mtq_smc, mtq_rate_damp; pointing: pid, lqr, smc, pid@bw2.5, pid@bw4	a run is scored against the case's requirements at run time channels are deleted after scoring (the manifest is kept)
11	assess	SILS	Python (tools/pipeline.py)	Per option: feasible on every seed? Each failing requirement classed; the best algorithm per option kept.	classes = power*: power; ake*: knowledge; propellant*: propellant; otherwise: performance violation = value / requirement - 1 (10 when not finite)	rank an option's variants by (not feasible, failing count, worst objective)

#	node	stage	runs in	what it does	parameters	rules
12	converge	design	Python (tools/pipeline.py)	The knob changes the failures call for; converged when nothing is left to change.	scale_bounds = 0.5, 4.0 scale_up = 1.5 scale_down = 0.75 improvement_needed = 0.05 fmr_lambda_bounds_kg_W = 0.01, 3.0 fmr_flow_sigma_min_m_s = 5e-05 gyro_grade_min = 0.1 max_iterations = 24	performance: fly every algorithm of the slot, then (fluid loop) a quieter flow sensor /4, then authority x1.5; undone when the violation does not fall by 5 % rate stability (fine class): gyro noise x0.3 down to x0.1; undone when it does not improve knowledge: fit the star tracker on a coarse product power, fluid loop: pump lambda x3 (not for RCS valve power); coils-only: coil authority x0.75; rotors: blocked (standby power is the floor) mass over budget, closest solution family: lighter gyro, one star-tracker head, lambda /3, fluid-loop momentum x0.75 -- each undone and closed if it breaks a mode or raises the violation by more than 5 % propellant: RCS x1.5 a part pushed up by one failure and down by another is frozen and reported as a conflict

#	node	stage	runs in	what it does	parameters	rules
13	select	selection	Python (tools/pipeline.py)	Scores every family (ours and the benchmarks) on the same modes and budget, then picks the minimum feasible solution; the benchmarks are ranked by the same rule as the comparison.	feasible = every mode passes with a usable option on every seed, and mass <= req.mass, volume <= req.vol rank_feasible = mass_kg, power_W, volume_L, simplicity rank_infeasible = gap_count, mass_kg select_role = solution compare_role = benchmark	selected = the feasible solution family first in rank_feasible order (least mass, then power, then volume) when no solution family is feasible the closest one is named with its gaps benchmark = the same ranking over the benchmark families, reported beside the selection
14	dispatch	dispatch	Rust (adcs dispatch)	The selected family's mission (detumble -> Sun acquisition -> nadir), its adcs-fswcfg/1 blob, the converged products; C and Rust checked bit-identical.	duration_orbits = detumble + acquisition + nadir	the C and Rust flight software must agree bit for bit
15	mc	verification	Rust engine	Monte Carlo of the dispatched mission with the case's dispersions (truth dispersed, flight software nominal).	runs = 12	dispersions as asils.campaign.draw
16	soft_oils	verification	Rust engine + QEMU (Cortex-M4F)	The dispatched mission with the flight software as firmware (C and Rust) beside its SILS run; exact instruction timing and bus latency.	obc_mhz = 168 cpi = 1.25 i2c_khz = 400 spi_mhz = 1 can_kbps = 1000	a latency >= the control step counts as an overrun a missing instruction count stops the run
17	report	report	Python	The design ledger per case and the V&V report (HTML and PDF)	—	every table is generated from the stored node outputs, never

4. Actuator catalogue (datasheets)

Reaction wheels, CMGs and VSCMGs are bought, so the benchmarks fly real products: the `select_rotor` node takes the lightest catalogue model that meets the case's per-unit momentum and torque (three wheels; a four-unit CMG pyramid, two units per axis). We design only the magnetorquers, the fluid loop and the RCS. Numbers are the vendors'; a model whose datasheet lacks momentum, torque, mass or steady power is listed but not selectable. Values the engine needs and datasheets omit (rotor inertia, friction, imbalance) are derived with the rules in each file (`tools/catalogue.py`, `docs/CATALOGUE.md`).

vendor	model	type	h [mNms]	torque [mNm]	mass [g]	steady [W]	peak [W]	size [mm]	selec
AAC Clyde Space	RW222-3	reaction_wheel	3	2	—	—	—	25 x 25 x 15	no: m power
AAC Clyde Space	RW222-6	reaction_wheel	6	2	—	—	—	25 x 25 x 15	no: m power
AAC Clyde Space	RW400-15	reaction_wheel	15	—	—	—	15	50 x 50 x 27	no: to mass_ power
AAC Clyde Space	RW400-30	reaction_wheel	30	—	—	—	15	50 x 50 x 27	no: to mass_ power
AAC Clyde Space	RW400-50	reaction_wheel	50	—	375	—	15	50 x 50 x 27	no: to power
CubeSpace	CubeWheel CW0017 (Gen2)	reaction_wheel	1.77	0.23	60	0.3	0.85	28 x 28 x 26	yes

vendor	model	type	h [mNms]	torque [mNm]	mass [g]	steady [W]	peak [W]	size [mm]	selec
CubeSpace	CubeWheel CW0057 (Gen2)	reaction_wheel	5.7	2	115	0.77	2.7	35 x 35 x 24	yes
CubeSpace	CubeWheel CW0162 (Gen2)	reaction_wheel	16.2	7	144	0.77	7.2	46 x 46 x 24	yes
CubeSpace	CubeWheel CW0500 (Gen2)	reaction_wheel	50	16	310	5.3	15	66 x 66 x 25	yes
CubeSpace	CubeWheel CW1200 (Gen2)	reaction_wheel	120	20	450	4.7	13	72 x 72 x 31	yes
CubeSpace	CubeWheel CW2500 (Gen2)	reaction_wheel	250	27	750	9	30	86 x 86 x 36	yes
CubeSpace	CubeWheel CW5000 (Gen2)	reaction_wheel	500	37	970	9	48	100 x 100 x 37	yes
Rocket Lab	RW3-0.06	reaction_wheel	60	20	226	0.5	23.4	77 x 65 x 38	yes
Rocket Lab	RW3-1.0	reaction_wheel	1000	50	970	2.4	46.7	150 x 140 x 42	yes
Rocket Lab	RW-0.003	reaction_wheel	3	1	50	—	—	33.5 x 33.5 x 17	no: power
Rocket Lab	RW-0.01	reaction_wheel	10	1	120	0.11	1.05	50 x 50 x 30	yes

vendor	model	type	h [mNms]	torque [mNm]	mass [g]	steady [W]	peak [W]	size [mm]	selec
Rocket Lab	RW-0.03	reaction_wheel	30	2	183	0.1	1.8	50 x 50 x 40	yes
Tensor Tech	ADCS400 (4 x RS100 variable- speed CMG pyramid)	cmg_cluster	10	1	300	1	1	0.2U+ tuna can per unit, >64 mm dia	yes
Tensor Tech	TensorCMG- 10m	cmg	13	100	—	—	—	—	no: m power

The vendor sites could not be reached from the build environment; numbers marked verified were read from the full datasheet PDFs (distributor mirror), the others from search excerpts of the cited page and should be confirmed with the vendor. Tensor Tech's ADCS400 figures are the whole integrated ADCS (CMGs, magnetorquer, sun sensors, gyro) as upper bounds, so the CMG benchmark's mass and power are conservative.

5. Requirements

ais_3u — 3U AIS satellite — coarse pointing 10 deg, SSO dawn-dusk 550 km

key	quantity	value	unit	source / status
req.ape	Absolute pointing error (APE)	10.0	Degree	stated by the user: the AIS antenna needs only coarse nadir pointing
req.ake	Absolute knowledge error (AKE)	5.0	Degree	UNCONFIRMED · plan reference: half the APE, so knowledge never dominates the budget
req.detumble	Detumble time	284.0	Minute	UNCONFIRMED · plan reference: three orbits (95.6 min each at 550 km) rounded as in the spec
req.sunacq	Sun-acquisition time in safe mode	95.0	Minute	UNCONFIRMED · plan reference: one orbit
req.mass	ADCS mass	1.6	Kilogram	set by the user (Agastya): the same ADCS budget in every 3U case, 1U of the 3U, 1.5-1.7 kg; 1.6 keeps 0.1 kg to the 1.7 ceiling
req.pavg	ADCS orbit-average power	0.5	Watt	UNCONFIRMED · plan reference: leaves the AIS receiver its share of a 3U power budget
req.ppk	ADCS peak power	1.5	Watt	UNCONFIRMED · plan reference: all coils at full current
req.vol	ADCS volume	1.0	Litre	set by the user (Agastya): 1U of the 3U is the ADCS
mission.epoch	Launch epoch	27.0	Year	UNCONFIRMED · adcs_ref_c1: years after J2000.0, early 2027
mission.spd	Slews per day	10.0	Count	UNCONFIRMED · adcs_ref_c1
mission.sangle	Reference slew angle	30.0	Degree	UNCONFIRMED · adcs_ref_c1
mission.w0	Initial tumble rate after separation	10.0	DegreePerSecond	UNCONFIRMED · adcs_ref_c1: tumble after separation
orbit.alt	Altitude	550.0	Kilometre	stated by the user (Agastya): 550 km LEO
orbit.inc	Inclination	97.593	Degree	sun-synchronous at 550 km: asils.orbit.sso_inclination(550)
orbit.ecc	Eccentricity	0.0	One	UNCONFIRMED · adcs_ref_c1
orbit.ltan	LTAN	6.0	Hour	stated by the user: SSO dawn-dusk (LTAN 06:00)
mass.m	Satellite mass	4.0	Kilogram	UNCONFIRMED · adcs_ref_c1
mass.imax	Largest principal inertia	0.042	KilogramSquareMetre	UNCONFIRMED · adcs_ref_c1: uniform 3U box, $m/12*(0.10^2+0.34^2)$
mass.iint	Intermediate principal inertia	0.042	KilogramSquareMetre	UNCONFIRMED · the same uniform-box model that gives mass.imax and mass.imin: the two long axes are equal

key	quantity	value	unit	source / status
mass.imin	Smallest principal inertia	0.0067	KilogramSquareMetre	UNCONFIRMED · adcs_ref_c1: uniform 3U box, $m/12*(0.10^2+0.10^2)$

ais_img_3u — 3U imaging satellite — fine pointing 0.01 deg, SSO 10:00 550 km

key	quantity	value	unit	source / status
req.ape	Absolute pointing error (APE)	0.01	Degree	stated by the user; 3-sigma. A flown 3U reference: MinXSS-1 (BCT XACT) measured 0.0042-0.0117 deg 3-sigma, axis dependent (arXiv:1706.06967)
req.ake	Absolute knowledge error (AKE)	0.005	Degree	UNCONFIRMED · plan reference: half the APE
req.rks	Rate stability	0.005	DegreePerSecond	relaxed by the user (Agastya) from 0.001: 18 arcsec/s; at 550 km that is 48 m/s of ground motion, 0.5 m of smear in a 10 ms exposure
req.slew	Reference slew time	60.0	Second	UNCONFIRMED · adcs_ref_c1: mission.sangle in 60 s, target to target
req.settle	Settling time after a slew	20.0	Second	UNCONFIRMED · plan reference: back inside the APE after a slew
req.detumble	Detumble time	284.0	Minute	UNCONFIRMED · plan reference: three orbits (95.6 min each at 550 km) rounded as in the spec
req.sunacq	Sun-acquisition time in safe mode	95.0	Minute	UNCONFIRMED · plan reference: one orbit
req.mass	ADCS mass	1.6	Kilogram	raised by the user (Agastya): 1.5-1.7 kg with 1U of the 3U for the ADCS; 1.6 keeps 0.1 kg to the 1.7 ceiling
req.pavg	ADCS orbit-average power	2.0	Watt	UNCONFIRMED · plan reference
req.ppk	ADCS peak power	4.0	Watt	UNCONFIRMED · plan reference
req.vol	ADCS volume	1.0	Litre	raised by the user (Agastya): 1U of the 3U is the ADCS
mission.epoch	Launch epoch	27.0	Year	UNCONFIRMED · adcs_ref_c1: years after J2000.0, early 2027
mission.spd	Slews per day	10.0	Count	UNCONFIRMED · adcs_ref_c1
mission.sangle	Reference slew angle	30.0	Degree	UNCONFIRMED · adcs_ref_c1
mission.w0	Initial tumble rate after separation	10.0	DegreePerSecond	UNCONFIRMED · adcs_ref_c1: tumble after separation
orbit.alt	Altitude	550.0	Kilometre	stated by the user (Agastya): 550 km LEO
orbit.inc	Inclination	97.593	Degree	sun-synchronous at 550 km: asils.orbit.sso_inclination(550)
orbit.ecc	Eccentricity	0.0	One	UNCONFIRMED · adcs_ref_c1

key	quantity	value	unit	source / status
orbit.ltan	LTAN	10.0	Hour	stated by the user: SSO 10:00 (read as LTAN 10:00; LTDN 22:00 gives the same plane and beta magnitude)
mass.m	Satellite mass	4.0	Kilogram	UNCONFIRMED · adcs_ref_c1
mass.imax	Largest principal inertia	0.042	KilogramSquareMetre	UNCONFIRMED · adcs_ref_c1: uniform 3U box, $m/12*(0.10^2+0.34^2)$
mass.iint	Intermediate principal inertia	0.042	KilogramSquareMetre	UNCONFIRMED · the same uniform-box model that gives mass.imax and mass.imin: the two long axes are equal
mass.imin	Smallest principal inertia	0.0067	KilogramSquareMetre	UNCONFIRMED · adcs_ref_c1: uniform 3U box, $m/12*(0.10^2+0.10^2)$

6. Design loop per case: every configuration compared

ais_3u: selected `mtq_fmr` — Magnetorquers + fluid momentum loop (feasible)

Converged after 4 iteration(s); knowledge class **coarse**; sensors: magnetometer, sun_sensors, gyro, gnss, earth_sensor. Demand on the POP orbit: peak disturbance 5.635e-07 N m (sun), stored momentum requirement 0.001466 N m s, torque requirement 5.757e-05 N m, weakest field 19.76 μ T, detumble momentum 0.00733 N m s.

iter	knobs	options feasible	selection	changes for the next iteration	blocked
1	laws as written	1/25	mtq (closest (not feasible))	nadir_pointing/fmr+mtq: power -> electromagnetic pump with more copper (lambda 0.1 -> 0.3 kg/W) nadir_pointing/mtq: fly every mtq_pointing algorithm sun_acquisition/mtq: fly every sun_acquisition algorithm sun_referencing/mtq: fly every mtq_pointing algorithm	16
2	pump λ 0.3 kg/W	4/25	mtq_fmr (feasible)	mtqp: authority x1 -> x1.5 (performance)	18
3	mtqp $\times$ 1.5, pump λ 0.3 kg/W	4/25	mtq_fmr (feasible)	mtqp: authority back to x1 (no improvement)	19
4	mtqp $\times$ 1, pump λ 0.3 kg/W	4/25	mtq_fmr (feasible)	—	19

Why the loop stopped (nothing left that a knob can change):

- detumble/rcs: power fails at the sized authority (rcs); the part's standby power is the floor
- mtqp: more authority did not reduce the performance violation (39.9 -> 41); kept at x1
- nadir_pointing/cmg+mtq: power fails at the sized authority (cmg); the part's standby power is the floor
- nadir_pointing/cmg+rcs: power fails at the sized authority (cmg); the part's standby power is the floor
- nadir_pointing/fmr+rcs: power is the thrusters' valve power (RCS dumping), not the pump
- nadir_pointing/rw+mtq: power fails at the sized authority (rw); the part's standby power is the floor
- nadir_pointing/rw+rcs: power fails at the sized authority (rw); the part's standby power is the floor
- nadir_pointing/vscmg+mtq: power fails at the sized authority (vscmg); the part's standby power is the floor
- nadir_pointing/vscmg+rcs: power fails at the sized authority (vscmg); the part's standby power is the floor
- sun_acquisition/cmg: power fails at the sized authority (cmg); the part's standby power is the floor
- sun_acquisition/rw: power fails at the sized authority (rw); the part's standby power is the floor
- sun_acquisition/vscmg: power fails at the sized authority (vscmg); the part's standby power is the floor
- sun_referencing/cmg+mtq: power fails at the sized authority (cmg); the part's standby power is the floor
- sun_referencing/cmg+rcs: power fails at the sized authority (cmg); the part's standby power is the floor
- sun_referencing/fmr+rcs: power is the thrusters' valve power (RCS dumping), not the pump
- sun_referencing/rw+mtq: power fails at the sized authority (rw); the part's standby power is the floor
- sun_referencing/rw+rcs: power fails at the sized authority (rw); the part's standby power is the floor
- sun_referencing/vscmg+mtq: power fails at the sized authority (vscmg); the part's standby power is the floor
- sun_referencing/vscmg+rcs: power fails at the sized authority (vscmg); the part's standby power is the floor

Every configuration, ours and the benchmarks, flown and scored the same way.

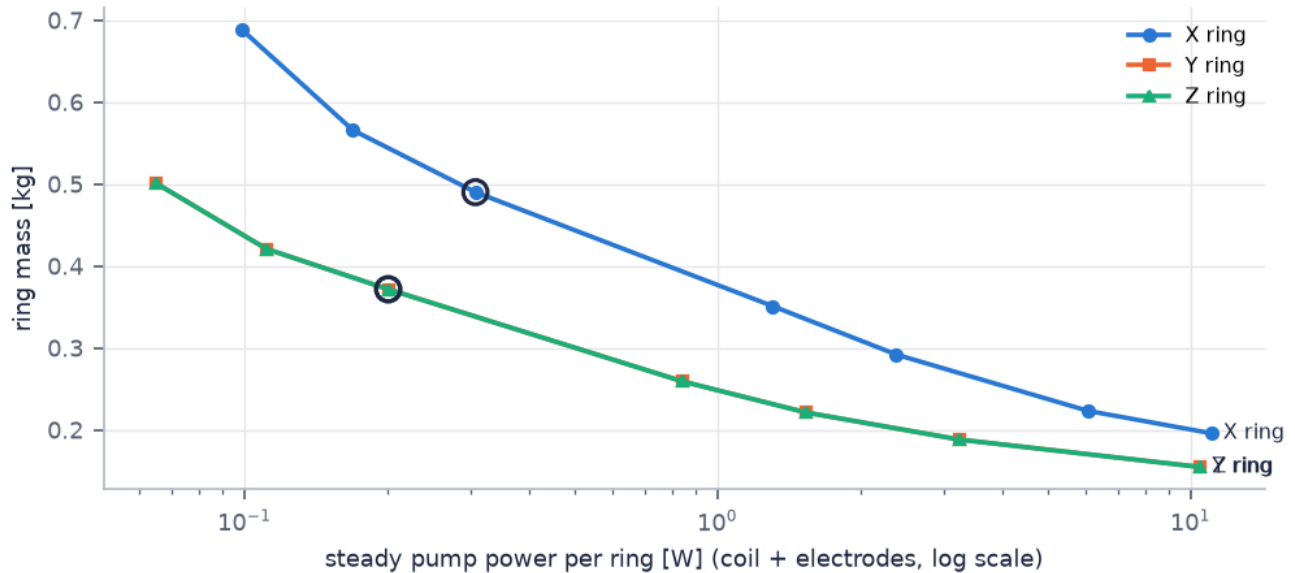
Selection rule (node `select`): least mass_kg, then power_W, then volume_L, then simplicity among feasible solution families. Selected: `mtq_fmr`; the benchmarks ranked by the same rule give `mtq_rw` (closest (not feasible)).

family	role	rank	feasible	mass [kg]	steady power [W]	volume [L]	momentum / thrust actuators	gaps
mtq_fmr	solution	1	pass	1.45	2.54	0.537	fluid loop, 3 rings (ours)	—
mtq	solution	—	FAIL	0.236	2.06	0.0183	coils only	sun_acquisiti sun_acquisiti (performance sun_angle_p (performance sun_referenc sun_ape_p99 (performance nadir_pointin ape_los_p99 (performance
mtq_fmr_rcs	solution	—	FAIL	1.96	2.59	1.52	Sized N2O cold-gas RCS, 6 couples ×1 fluid loop, 3 rings (ours)	budget: mass 1.96 > 1.6; b volume_L 1.5
mtq_rw	benchmark	—	FAIL	0.393	2.73	0.0703	CubeSpace CubeWheel CW0017 ×3	sun_acquisiti power_mean (power); sun_referenc power_mean (power); nadir_pointin power_mean (power)
mtq_rw_rcs	benchmark	—	FAIL	0.911	2.78	1.05	CubeSpace CubeWheel CW0017 ×3 Sized N2O cold-gas RCS, 6 couples ×1	sun_acquisiti power_mean (power); sun_referenc power_mean (power); nadir_pointin power_mean (power); bud volume_L 1.0
mtq_cmj	benchmark	—	FAIL	1.41	5.83	0.809	Tensor Tech ADCS400 ×4	sun_acquisiti power_mean (power); sun_referenc power_mean (power); nadir_pointin power_mean (power)

family	role	rank	feasible	mass [kg]	steady power [W]	volume [L]	momentum / thrust actuators	gaps
mtq_vscmg	benchmark	—	FAIL	1.41	5.83	0.809	Tensor Tech ADCS400 ×4	sun_acquisition power_mean (power); sun_referencing power_mean (power); nadir_pointing power_mean (power)
mtq_cmg_rcs	benchmark	—	FAIL	1.93	5.88	1.79	Sized N2O cold-gas RCS, 6 couples ×1 Tensor Tech ADCS400 ×4	sun_acquisition power_mean (power); sun_referencing power_mean (power); nadir_pointing power_mean (power); budget: volume 1.79 > 1
mtq_vscmg_rcs	benchmark	—	FAIL	1.93	5.88	1.79	Sized N2O cold-gas RCS, 6 couples ×1 Tensor Tech ADCS400 ×4	sun_acquisition power_mean (power); sun_referencing power_mean (power); nadir_pointing power_mean (power); budget: volume 1.79 > 1

Selected method per mission mode:

mode	option	feasible	objective (worst seed)	algorithms
detumble	mtq	pass	48.74 detumble_time	attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, sun_acquisition=sunspin_l1l2
sun_acquisition	fmr	pass	2.292 sun_acquisition_time	allocation=idmas_split, attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, pointing=pid, sun_acquisition=sunspin_l1l2
sun_referencing	fmr+mtq	pass	0.2513 sun_ape_p9973	allocation=idmas_split, attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, pointing=pid, sun_acquisition=sunspin_l1l2
nadir_pointing	fmr+mtq	pass	0.1514 ape_los_p9973	allocation=idmas_split, attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, pointing=pid, sun_acquisition=sunspin_l1l2



Electromagnetic pump design, ais_3u: each ring's mass against steady power as the mass/power rate λ runs from 0.003 to 3 kg/W; circles mark the converged designs ($\lambda = 0.3$ kg/W).

Monte Carlo of the dispatched mission (12 runs; inertia, residual_dipole, cm_offset, solar_flux, kp, accommodation, reflectivity, arg_lat_deg):

metric	req	mean $\pm$ std	range	pass rate
detumble_time	284	58.22 $\pm$ 22.1	[9.858, 94.17]	100 %
ape_los_p9973	10	0.2426 $\pm$ 0.0731	[0.1603, 0.3825]	100 %
ake_los_p9973	5	0.2393 $\pm$ 0.0732	[0.1574, 0.3818]	100 %
power_mean	0.5	0.3186 $\pm$ 0.0923	[0.2113, 0.493]	100 %
power_peak	—	0.859 $\pm$ 0.0129	[0.8384, 0.8847]	—
propellant	—	0 $\pm$ 0	[0, 0]	—

The dispatched mission in SILS and in soft OILS (flight software as Cortex-M4F firmware, C and Rust builds):

metric	req	SILS	soft OILS, C	soft OILS, Rust
detumble_time	284	55.64 pass	55.71 pass	55.71 pass
ape_los_p9973	10	0.1516 pass	0.1217 pass	0.1274 pass
ake_los_p9973	5	0.15 pass	0.1184 pass	0.1234 pass
power_mean	0.5	0.233 pass	0.2175 pass	0.2189 pass
power_peak	—	0.8385 —	0.85 —	0.8508 —
propellant	—	0 —	0 —	0 —

ais_img_3u: selected `mtq_fmr` — Magnetorquers + fluid momentum loop (feasible)

Converged after 7 iteration(s); knowledge class **fine**; sensors: star_tracker, magnetometer, sun_sensors, gyro, gnss, earth_sensor. Demand on the POP orbit: peak disturbance 4.367e-07 N m (sun), stored momentum requirement 0.001552 N m s, torque requirement 5.757e-05 N m, weakest field 23.63 μ T, detumble momentum 0.00733 N m s.

iter	knobs	options feasible	selection	changes for the next iteration	blocked
1	laws as written	9/25	mtq_fmr (closest (not feasible))	nadir_pointing/fmr+mtq: fly every pointing algorithm nadir_pointing/fmr+rscs: fly every pointing algorithm nadir_pointing/mtq: fly every mtq_pointing algorithm sun_acquisition/mtq: fly every sun_acquisition algorithm sun_referencing/mtq: fly every mtq_pointing algorithm	13
2	laws as written	9/25	mtq_fmr (closest (not feasible))	rate stability: gyro noise x1 -> x0.3 (fibre-optic class) mtqp: authority x1 -> x1.5 (performance)	14
3	mtqp ×1.5, gyro noise x0.3	9/25	mtq_fmr (closest (not feasible))	gyro grade back to x1: rate-stability violation 13.8 -> 13.8 mtqp: authority back to x1 (no improvement) nadir_pointing/fmr+mtq: fluid-loop flow sensor 2 -> 0.5 mm/s (1 sigma)	15
4	mtqp ×1, flow sensor 0.5 mm/s	10/25	mtq_fmr (closest (not feasible))	mass (mtq_fmr): one star-tracker head instead of two	14
5	mtqp ×1, 1 ST head, flow sensor 0.5 mm/s	7/25	mtq_fmr (closest (not feasible))	nadir_pointing/cmg+mtq: fly every pointing algorithm nadir_pointing/cmg+rscs: fly every pointing algorithm nadir_pointing/fmr+mtq: fluid-loop flow sensor 0.5 -> 0.125 mm/s (1 sigma) nadir_pointing/rw+mtq: fly every pointing algorithm nadir_pointing/rw+rscs: fly every pointing algorithm nadir_pointing/vscmg+mtq: fly every pointing algorithm nadir_pointing/vscmg+rscs: fly every pointing algorithm mass lever st_heads undone: it broke nadir_pointing/fmr+mtq, requirement violation 0 -> 41.1	17
6	mtqp ×1, flow sensor 0.125 mm/s	12/25	mtq_fmr (closest (not feasible))	mass (mtq_fmr): lighter pump, less copper (lambda 0.1 -> 0.0333333 kg/W)	12
7	mtqp ×1, pump λ 0.0333333 kg/W, flow sensor 0.125 mm/s	11/25	mtq_fmr (feasible)	—	13

Why the loop stopped (nothing left that a knob can change):

- mtqp: more authority did not reduce the performance violation ($1.33e+04 \rightarrow 1.34e+04$); kept at x1
- nadir_pointing/cmg+mtq: power fails at the sized authority (cmg); the part's standby power is the floor
- nadir_pointing/cmg+rscs: power fails at the sized authority (cmg); the part's standby power is the floor
- nadir_pointing/mtq: knowledge fails with the star tracker fitted
- nadir_pointing/vscmg+mtq: power fails at the sized authority (vscmg); the part's standby power is the floor
- nadir_pointing/vscmg+rscs: power fails at the sized authority (vscmg); the part's standby power is the floor
- sun_acquisition/cmg: power fails at the sized authority (cmg); the part's standby power is the floor
- sun_acquisition/vscmg: power fails at the sized authority (vscmg); the part's standby power is the floor
- sun_referencing/cmg+mtq: power fails at the sized authority (cmg); the part's standby power is the floor

- sun_referencing/cmg+rcs: power fails at the sized authority (cmg); the part's standby power is the floor
- sun_referencing/fmr+rcs: power is the thrusters' valve power (RCS dumping), not the pump
- sun_referencing/vscmg+mtq: power fails at the sized authority (vscmg); the part's standby power is the floor
- sun_referencing/vscmg+rcs: power fails at the sized authority (vscmg); the part's standby power is the floor

Every configuration, ours and the benchmarks, flown and scored the same way.

Selection rule (node `select`): least mass_kg, then power_W, then volume_L, then

simplicity among feasible solution families. Selected: `mtq_fmr`; the benchmarks

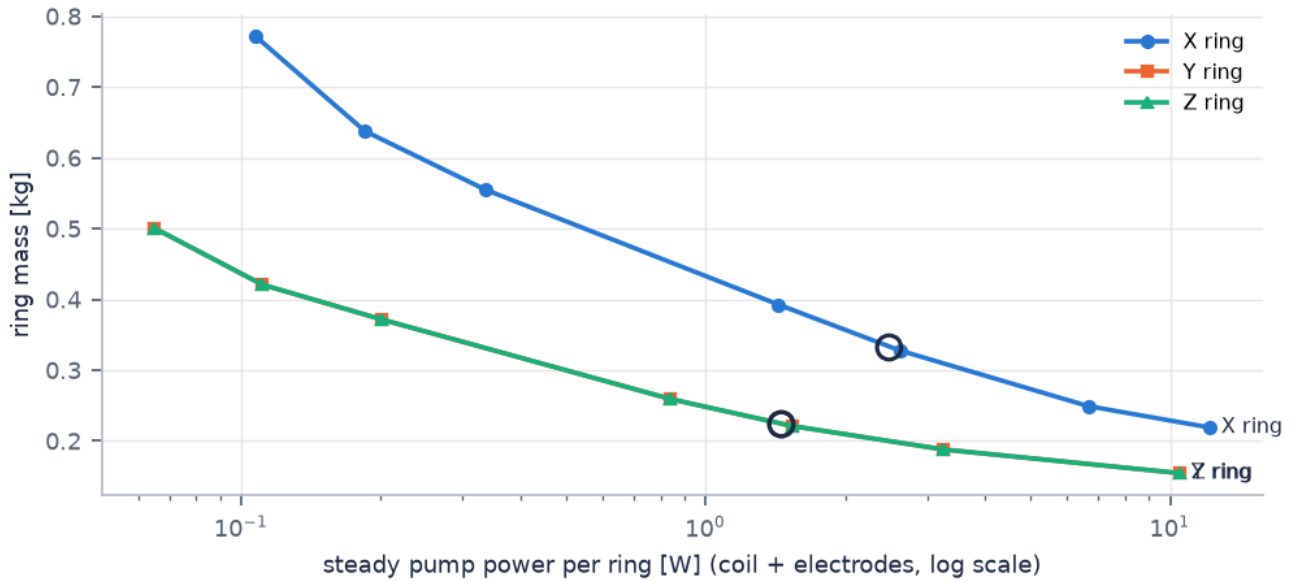
ranked by the same rule give `mtq_rw` (feasible).

family	role	rank	feasible	mass [kg]	steady power [W]	volume [L]	momentum / thrust actuators	gaps
<code>mtq_fmr</code>	solution	1	pass	1.6	9.16	0.534	fluid loop, 3 rings (ours)	—
<code>mtq</code>	solution	—	FAIL	0.83	3.9	0.0118	coils only	sun_acquisiti sun_acquisiti (performance sun_angle_p (performance sun_referenc sun_ape_p99 (performance nadir_pointin ape_los_p99 (performance ake_los_p99 (knowledge), rate_stability (performance
<code>mtq_fmr_rcs</code>	solution	—	FAIL	2.06	9.21	1.38	Sized N2O cold-gas RCS, 6 couples ×1 fluid loop, 3 rings (ours)	budget: mass 2.06 > 1.6; b volume_L 1.3
<code>mtq_rw</code>	benchmark	1	pass	0.996	4.66	0.0676	CubeSpace CubeWheel CW0017 ×3	—
<code>mtq_rw_rcs</code>	benchmark	2	pass	1.46	4.71	0.909	CubeSpace CubeWheel CW0017 ×3 Sized N2O cold-gas RCS, 6 couples ×1	—
<code>mtq_cmg</code>	benchmark	—	FAIL	2.02	7.76	0.806	Tensor Tech ADCS400 ×4	sun_acquisiti power_mean (power); sun_referenc power_mean (power); nadir_pointin power_mean (power); bud mass_kg 2.0

family	role	rank	feasible	mass [kg]	steady power [W]	volume [L]	momentum / thrust actuators	gaps
mtq_vscmg	benchmark	—	FAIL	2.02	7.76	0.806	Tensor Tech ADCS400 ×4	sun_acquisiti power_mean (power); sun_referenc power_mean (power); nadir_pointin power_mean (power); bud mass_kg 2.0
mtq_cmrg_rcs	benchmark	—	FAIL	2.48	7.81	1.65	Sized N2O cold-gas RCS, 6 couples ×1 Tensor Tech ADCS400 ×4	sun_acquisiti power_mean (power); sun_referenc power_mean (power); nadir_pointin power_mean (power); bud mass_kg 2.4 budget: volur 1.65 > 1
mtq_vscmg_rcs	benchmark	—	FAIL	2.48	7.81	1.65	Sized N2O cold-gas RCS, 6 couples ×1 Tensor Tech ADCS400 ×4	sun_acquisiti power_mean (power); sun_referenc power_mean (power); nadir_pointin power_mean (power); bud mass_kg 2.4 budget: volur 1.65 > 1

Selected method per mission mode:

mode	option	feasible	objective (worst seed)	algorithms
detumble	mtq	pass	63.54 detumble_time	attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, sun_acquisition=sunspin_l1l2
sun_acquisition	fmr	pass	2.275 sun_acquisition_time	allocation=idmas_split, attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, pointing=pid, sun_acquisition=sunspin_l1l2
sun_referencing	fmr+mtq	pass	0.008977 sun_ape_p9973	allocation=idmas_split, attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, pointing=pid, sun_acquisition=sunspin_l1l2
nadir_pointing	fmr+mtq	pass	0.003535 ape_los_p9973	allocation=idmas_split, attitude=mekf, detumble=bdot_gyro, mtq_pointing=mtq_pd, pointing=pid, sun_acquisition=sunspin_l1l2



Electromagnetic pump design, ais_img_3u: each ring's mass against steady power as the mass/power rate λ runs from 0.003 to 3 kg/W; circles mark the converged designs ($\lambda = 0.0333333$ kg/W).

Monte Carlo of the dispatched mission (12 runs; inertia, residual_dipole, cm_offset, solar_flux, accommodation, reflectivity, arg_lat_deg):

metric	req	mean ± std	range	pass rate
detumble_time	284	73.64 ± 15.5	[44.49, 92.39]	100 %
ape_los_p9973	0.01	0.003414 ± 0.000333	[0.002923, 0.004088]	100 %
ake_los_p9973	0.005	0.002662 ± 0.000348	[0.001964, 0.003348]	100 %
power_mean	2	1.497 ± 0.802	[0.239, 2.864]	75 %
power_peak	—	5.499 ± 0.00991	[5.483, 5.514]	—
propellant	—	0 ± 0	[0, 0]	—

The dispatched mission in SILS and in soft OILS (flight software as Cortex-M4F firmware, C and Rust builds):

metric	req	SILS	soft OILS, C	soft OILS, Rust
detumble_time	284	81.91 pass	82.81 pass	82.76 pass
ape_los_p9973	0.01	0.003645 pass	0.003707 pass	0.003734 pass
ake_los_p9973	0.005	0.002948 pass	0.002961 pass	0.002959 pass
power_mean	2	1.049 pass	0.9794 pass	1.222 pass
power_peak	—	5.489 —	5.499 —	5.49 —
propellant	—	0 —	0 —	0 —

7. SILS scenario verification

Every scenario on the Rust engine with the C flight software behind the byte HAL (POP in the loop). 25 of 40 scenarios meet every requirement they carry; the failures are findings of the design (section 10), not engine errors: the same verdicts come out of the MATLAB twin (section 7).

scenario	what	product	duration [s]	requirement metrics: value / req
agile_slew_cmgy	3U imaging — (cmgy): 90 deg pitch in 15 s	TRN-P-3U-CMG	600	ape_los_on_target_p9973 0.004544 / 0.01 pass
agile_slew_fmr_rcs	3U imaging — (fmr_rcs): 90 deg pitch in 15 s	TRN-P-3U-FMR-RCS	600	ape_los_on_target_p9973 0.006614 / 0.01 pass
agile_slew_img	3U imaging — (img): 90 deg pitch in 15 s	TRN-P-3U-IMG	600	ape_los_on_target_p9973 0.008686 / 0.01 pass
agile_slew_rw_rcs	3U imaging — (rw_rcs): 90 deg pitch in 15 s	TRN-P-3U-RW-RCS	600	ape_los_on_target_p9973 0.008702 / 0.01 pass
agile_slew_vscmg	3U imaging — (vscmg): 90 deg pitch in 15 s	TRN-P-3U-VSCMG	600	ape_los_on_target_p9973 0.01008 / 0.01 FAIL
detumble_ais	3U AIS — B-dot detumble from a 10 deg/s tumble (coils only)	TRN-P-3U-AIS	1.722e+04	detumble_time 37.79 / 284 pass power_mean 0.0217 / 0.5 pass power_peak 0.5502 / 1.5 pass
detumble_ais_bangbang	3U AIS — detumble, B-dot law bangbang	TRN-P-3U-AIS	1.722e+04	detumble_time 54.02 / 284 pass power_mean 0.4366 / 0.5 pass power_peak 0.717 / 1.5 pass
detumble_ais_mag	3U AIS — detumble, B-dot law mag	TRN-P-3U-AIS	1.722e+04	detumble_time 41.16 / 284 pass power_mean 0.07839 / 0.5 pass power_peak 0.6377 / 1.5 pass
detumble_img	3U imaging — B-dot detumble from a 10 deg/s tumble	TRN-P-3U-IMG	1.722e+04	detumble_time 47.54 / 284 pass
fault_coil_ais	3U AIS — X coil fails during detumble	TRN-P-3U-AIS	1.722e+04	detumble_time 82.17 / 284 pass power_mean 0.01821 / 0.5 pass power_peak 0.5502 / 1.5 pass
fault_gimbal_cmgy	3U imaging — one CMG gimbal sticks during fine hold	TRN-P-3U-CMG	5740	ape_los_p9973 0.005509 / 0.01 pass ake_los_p9973 0.002624 / 0.005 pass rate_stability_p9973 0.002963 / 0.005 pass power_mean 1.61 / 2 pass
fault_gyro_ais	3U AIS — gyro bias step of 0.1 deg/s during nadir hold	TRN-P-3U-AIS	1.722e+04	ape_los_last_orbit 52.78 / 10 FAIL ake_los_last_orbit 41.92 / 5 FAIL power_mean 0.01056 / 0.5 pass

scenario	what	product	duration [s]	requirement metrics: value / req
fault_st_img	3U imaging — star-tracker head 2 fails during fine hold	TRN-P-3U-IMG	5740	ape_los_p9973 0.1022 / 0.01 FAIL ake_los_p9973 0.1093 / 0.005 FAIL rate_stability_p9973 0.003771 / 0.005 pass power_mean 1.557 / 2 pass
fault_wheel_img	3U imaging — reaction wheel 1 fails during fine hold (FDIR + coils)	TRN-P-3U-IMG	5740	ape_los_p9973 28.62 / 0.01 FAIL ake_los_p9973 0.002858 / 0.005 pass rate_stability_p9973 0.6244 / 0.005 FAIL power_mean 1.213 / 2 pass
fine_hold_cmrg	3U imaging — fine nadir hold, coils + 4 SGCMG pyramid	TRN-P-3U-CMG	5740	ape_los_p9973 0.005021 / 0.01 pass ake_los_p9973 0.002626 / 0.005 pass rate_stability_p9973 0.002437 / 0.005 pass power_mean 1.61 / 2 pass
fine_hold_fmr	3U imaging — fine nadir hold, coils + fluid momentum rings (IDMAS split)	TRN-P-3U-FMR	5740	ape_los_p9973 0.005689 / 0.01 pass ake_los_p9973 0.002631 / 0.005 pass rate_stability_p9973 0.004729 / 0.005 pass power_mean 0.08406 / 2 pass
fine_hold_fmr_rcs	3U imaging — fine nadir hold, coils + fluid rings + cold-gas RCS	TRN-P-3U-FMR-RCS	5740	ape_los_p9973 0.005689 / 0.01 pass ake_los_p9973 0.002631 / 0.005 pass rate_stability_p9973 0.004729 / 0.005 pass power_mean 0.08406 / 2 pass
fine_hold_img	3U imaging — fine nadir hold, reaction wheels + star tracker (0.01 deg)	TRN-P-3U-IMG	5740	ape_los_p9973 0.01049 / 0.01 FAIL ake_los_p9973 0.002608 / 0.005 pass rate_stability_p9973 0.003552 / 0.005 pass power_mean 1.557 / 2 pass
fine_hold_img_lqr	3U imaging — fine nadir hold, LQR controller	TRN-P-3U-IMG	5740	ape_los_p9973 0.004156 / 0.01 pass ake_los_p9973 0.002652 / 0.005 pass rate_stability_p9973 0.006704 / 0.005 FAIL power_mean 1.558 / 2 pass
fine_hold_img_smc	3U imaging — fine nadir hold, SMC controller	TRN-P-3U-IMG	5740	ape_los_p9973 0.01945 / 0.01 FAIL ake_los_p9973 0.002649 / 0.005 pass rate_stability_p9973 0.003482 / 0.005 pass power_mean 1.557 / 2 pass

scenario	what	product	duration [s]	requirement metrics: value / req
<code>fine_hold_rw_rcs</code>	3U imaging — fine nadir hold, coils + reaction wheels + cold-gas RCS	TRN-P-3U-RW-RCS	5740	ape_los_p9973 0.03388 / 0.01 FAIL ake_los_p9973 0.002602 / 0.005 pass rate_stability_p9973 0.005641 / 0.005 FAIL power_mean 1.53 / 2 pass
<code>fine_hold_vscmg</code>	3U imaging — fine nadir hold, coils + 4 VSCMG pyramid	TRN-P-3U-VSCMG	5740	ape_los_p9973 0.00828 / 0.01 pass ake_los_p9973 0.002636 / 0.005 pass rate_stability_p9973 0.004762 / 0.005 pass power_mean 2.014 / 2 FAIL
<code>mission_ais</code>	3U AIS — integrated chain: detumble then nadir hold (auto mode switch)	TRN-P-3U-AIS	2.87e+04	detumble_time 37.79 / 284 pass ape_los_last_orbit 8.935 / 10 pass ake_los_last_orbit 1.188 / 5 pass
<code>mission_cmg</code>	3U imaging — tumble, detumble, fine hold: coils + 4 SGCMG pyramid	TRN-P-3U-CMG	1.722e+04	detumble_time 57.01 / 284 pass ape_los_last_half_orbit_p9973 0.005241 / 0.01 pass
<code>mission_fmr</code>	3U imaging — tumble, detumble, fine hold: coils + fluid momentum rings (IDMAS split)	TRN-P-3U-FMR	1.722e+04	detumble_time 61.33 / 284 pass ape_los_last_half_orbit_p9973 0.005954 / 0.01 pass
<code>mission_fmr_rcs</code>	3U imaging — tumble, detumble, fine hold: coils + fluid rings + cold-gas RCS	TRN-P-3U-FMR-RCS	1.722e+04	detumble_time 61.33 / 284 pass ape_los_last_half_orbit_p9973 0.005954 / 0.01 pass
<code>mission_img</code>	3U imaging — integrated chain: detumble then fine nadir hold	TRN-P-3U-IMG	1.722e+04	detumble_time 51.01 / 284 pass ape_los_last_half_orbit_p9973 0.01066 / 0.01 FAIL
<code>mission_rw_rcs</code>	3U imaging — tumble, detumble, fine hold: coils + reaction wheels + cold-gas RCS	TRN-P-3U-RW-RCS	1.722e+04	detumble_time 51.01 / 284 pass ape_los_last_half_orbit_p9973 0.007564 / 0.01 pass
<code>mission_vscmg</code>	3U imaging — tumble, detumble, fine hold: coils + 4 VSCMG pyramid	TRN-P-3U-VSCMG	1.722e+04	detumble_time 57.02 / 284 pass ape_los_last_half_orbit_p9973 0.009595 / 0.01 pass
<code>nadir_hold_ais</code>	3U AIS — nadir hold with magnetorquers only (10 deg requirement)	TRN-P-3U-AIS	1.722e+04	ape_los_last_orbit 12.64 / 10 FAIL ake_los_last_orbit 5.491 / 5 FAIL power_mean 0.01054 / 0.5 pass
<code>nadir_hold_ais_css</code>	3U AIS — nadir hold with coarse cosine sun sensors (albedo)	TRN-P-3U-AIS-CSS	1.722e+04	ape_los_last_orbit 39.44 / 10 FAIL ake_los_last_orbit 39.22 / 5 FAIL power_mean 0.01064 / 0.5 pass
<code>slew_cmg</code>	3U imaging — 30 deg target slew in 60 s, coils + 4 SGCMG pyramid	TRN-P-3U-CMG	600	settle_time_after_slew 12.4 / 20 pass ape_los_on_target_p9973 0.004233 / 0.01 pass
<code>slew_fmr</code>	3U imaging — 30 deg target slew in 60 s, coils + fluid momentum rings (IDMAS split)	TRN-P-3U-FMR	600	settle_time_after_slew 12.9 / 20 pass ape_los_on_target_p9973 0.006529 / 0.01 pass

scenario	what	product	duration [s]	requirement metrics: value / req
slew_fmr_rcs	3U imaging — 30 deg target slew in 60 s, coils + fluid rings + cold-gas RCS	TRN-P- 3U-FMR- RCS	600	settle_time_after_slew 12.9 / 20 pass ape_los_on_target_p9973 0.006529 / 0.01 pass
slew_img	3U imaging — 30 deg target slew in 60 s and settle	TRN-P- 3U-IMG	600	settle_time_after_slew 12.1 / 20 pass ape_los_on_target_p9973 0.007925 / 0.01 pass
slew_img_lqr	3U imaging — 30 deg slew, LQR controller	TRN-P- 3U-IMG	600	settle_time_after_slew 0 / 20 pass ape_los_on_target_p9973 0.004439 / 0.01 pass
slew_img_smc	3U imaging — 30 deg slew, SMC controller	TRN-P- 3U-IMG	600	settle_time_after_slew — / 20 FAIL ape_los_on_target_p9973 0.0178 / 0.01 FAIL
slew_rw_rcs	3U imaging — 30 deg target slew in 60 s, coils + reaction wheels + cold-gas RCS	TRN-P- 3U-RW- RCS	600	settle_time_after_slew 12.1 / 20 pass ape_los_on_target_p9973 0.007922 / 0.01 pass
slew_vscmg	3U imaging — 30 deg target slew in 60 s, coils + 4 VSCMG pyramid	TRN-P- 3U- VSCMG	600	settle_time_after_slew 13.2 / 20 pass ape_los_on_target_p9973 0.01083 / 0.01 FAIL
sun_spin_ais	3U AIS — coils-only Sun acquisition: detumble, spin- up, Sun-pointing spin	TRN-P- 3U-AIS	1.722e+04	sun_spin_entry 142.4 / 95 FAIL sun_angle_last_orbit 93.84 / 20 FAIL spin_rate_error_last_orbit 1.116 / 0.5 FAIL sun_spin_share_last_orbit 13.75 / 90 FAIL power_mean 0.45 / 0.5 pass

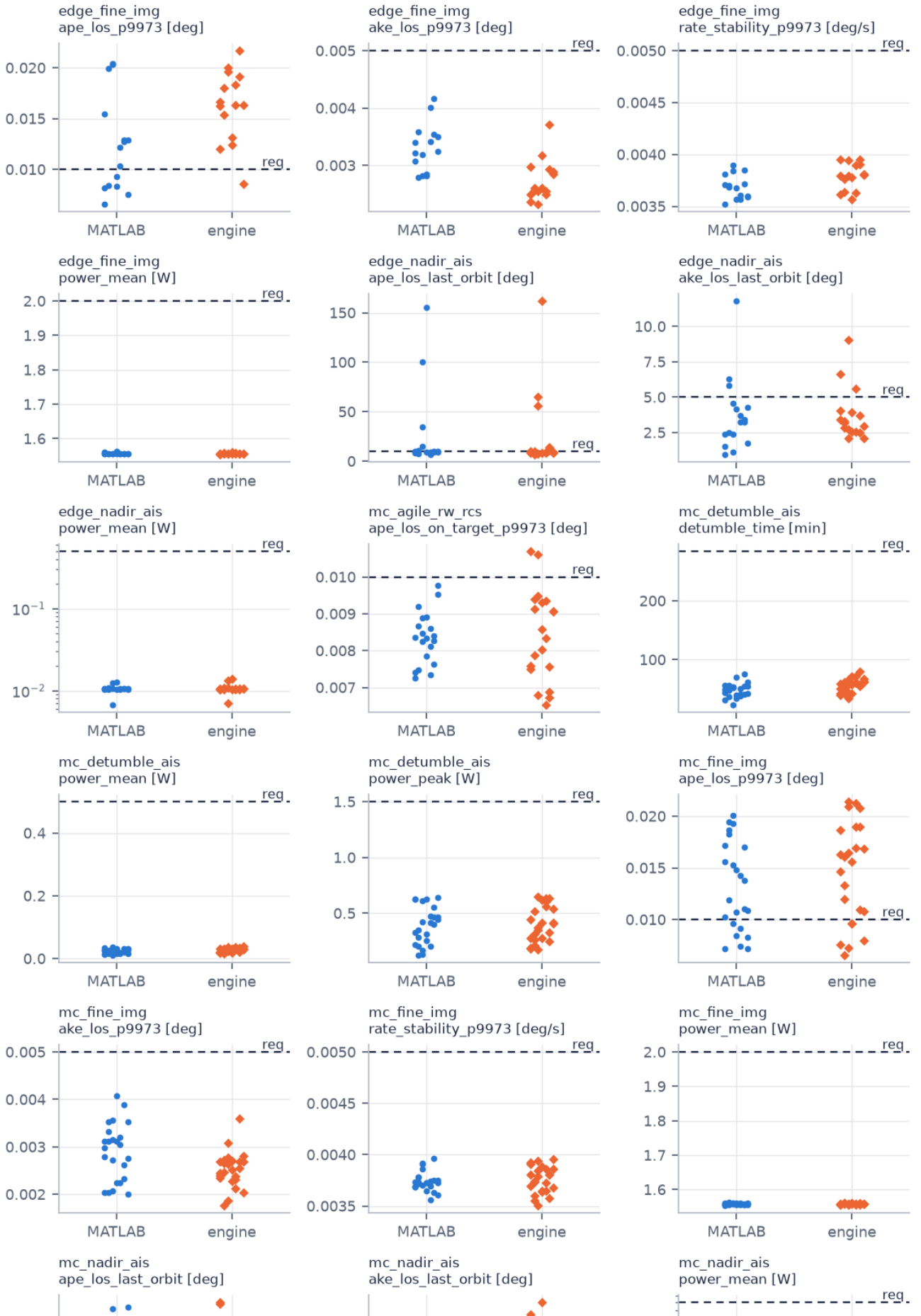
8. Monte Carlo and edge campaigns vs the MATLAB twin

Every campaign of `campaigns/*.toml` flown on the Rust engine with the draws of `asils.campaign.draw` (same dispersions, bounds, run count, per-run seeds), next to the MATLAB twin's campaign. Monte Carlo realisations differ (different random streams) so distributions are compared; edge runs are compared run by run. The engine's flight software keeps the nominal inertia while the plant is dispersed; the MATLAB twin's laws read the dispersed inertia.

campaign	type	metric	req	MATLAB mean $\pm$ std	MATLAB pass	engine mean $\pm$ std	engine pass
edge_fine_img	edge	ape_los_p9973	0.01	0.01236 $\pm$ 0.00475	40 %	0.01625 $\pm$ 0.00351	7 %
edge_fine_img	edge	ake_los_p9973	0.005	0.003307 $\pm$ 0.000418	100 %	0.002739 $\pm$ 0.000362	100 %
edge_fine_img	edge	rate_stability_p9973	0.005	0.003689 $\pm$ 0.000116	100 %	0.003788 $\pm$ 0.00013	100 %
edge_fine_img	edge	power_mean	2	1.558 $\pm$ 0.00216	100 %	1.557 $\pm$ 0.00112	100 %
edge_nadir_ais	edge	ape_los_last_orbit	10	24.91 $\pm$ 40.4	59 %	24.19 $\pm$ 39.4	65 %
edge_nadir_ais	edge	ake_los_last_orbit	5	3.712 $\pm$ 2.58	82 %	3.713 $\pm$ 1.82	82 %
edge_nadir_ais	edge	power_mean	0.5	0.01057 $\pm$ 0.00121	100 %	0.01069 $\pm$ 0.00144	100 %
mc_agile_rw_rcs	montecarlo	ape_los_on_target_p9973	0.01	0.008337 $\pm$ 0.000712	100 %	0.008427 $\pm$ 0.00124	90 %
mc_detumble_ais	montecarlo	detumble_time	284	48.5 $\pm$ 12	100 %	56.29 $\pm$ 11.4	100 %
mc_detumble_ais	montecarlo	power_mean	0.5	0.02267 $\pm$ 0.00701	100 %	0.02764 $\pm$ 0.00684	100 %
mc_detumble_ais	montecarlo	power_peak	1.5	0.3902 $\pm$ 0.169	100 %	0.402 $\pm$ 0.16	100 %
mc_fine_img	montecarlo	ape_los_p9973	0.01	0.01315 $\pm$ 0.00434	29 %	0.01456 $\pm$ 0.00486	25 %
mc_fine_img	montecarlo	ake_los_p9973	0.005	0.002894 $\pm$ 0.000605	100 %	0.002523 $\pm$ 0.000383	100 %
mc_fine_img	montecarlo	rate_stability_p9973	0.005	0.003738 $\pm$ 9.57e-05	100 %	0.003755 $\pm$ 0.00013	100 %
mc_fine_img	montecarlo	power_mean	2	1.558 $\pm$ 0.0021	100 %	1.558 $\pm$ 0.00149	100 %
mc_nadir_ais	montecarlo	ape_los_last_orbit	10	89.69 $\pm$ 38.5	0 %	92.98 $\pm$ 42.3	0 %
mc_nadir_ais	montecarlo	ake_los_last_orbit	5	3.3 $\pm$ 1.73	75 %	2.829 $\pm$ 1.69	88 %
mc_nadir_ais	montecarlo	power_mean	0.5	0.01145 $\pm$ 0.0025	100 %	0.01004 $\pm$ 0.0029	100 %
mc_slew_cmgy	montecarlo	settle_time_after_slew	20	13.14 $\pm$ 0.46	100 %	13.03 $\pm$ 0.785	100 %
mc_slew_cmgy	montecarlo	ape_los_on_target_p9973	0.01	0.003931 $\pm$ 0.000635	100 %	0.003761 $\pm$ 0.000349	100 %

campaign	type	metric	req	MATLAB mean ± std	MATLAB pass	engine mean ± std	engine pass
mc_slew_img	montecarlo	settle_time_after_slew	20	13.03 ± 0.959	100 %	13.84 ± 4.76	98 %
mc_slew_img	montecarlo	ape_los_on_target_p9973	0.01	0.00815 ± 0.00115	92 %	0.007886 ± 0.001	95 %

• MATLAB twin ♦ Rust engine



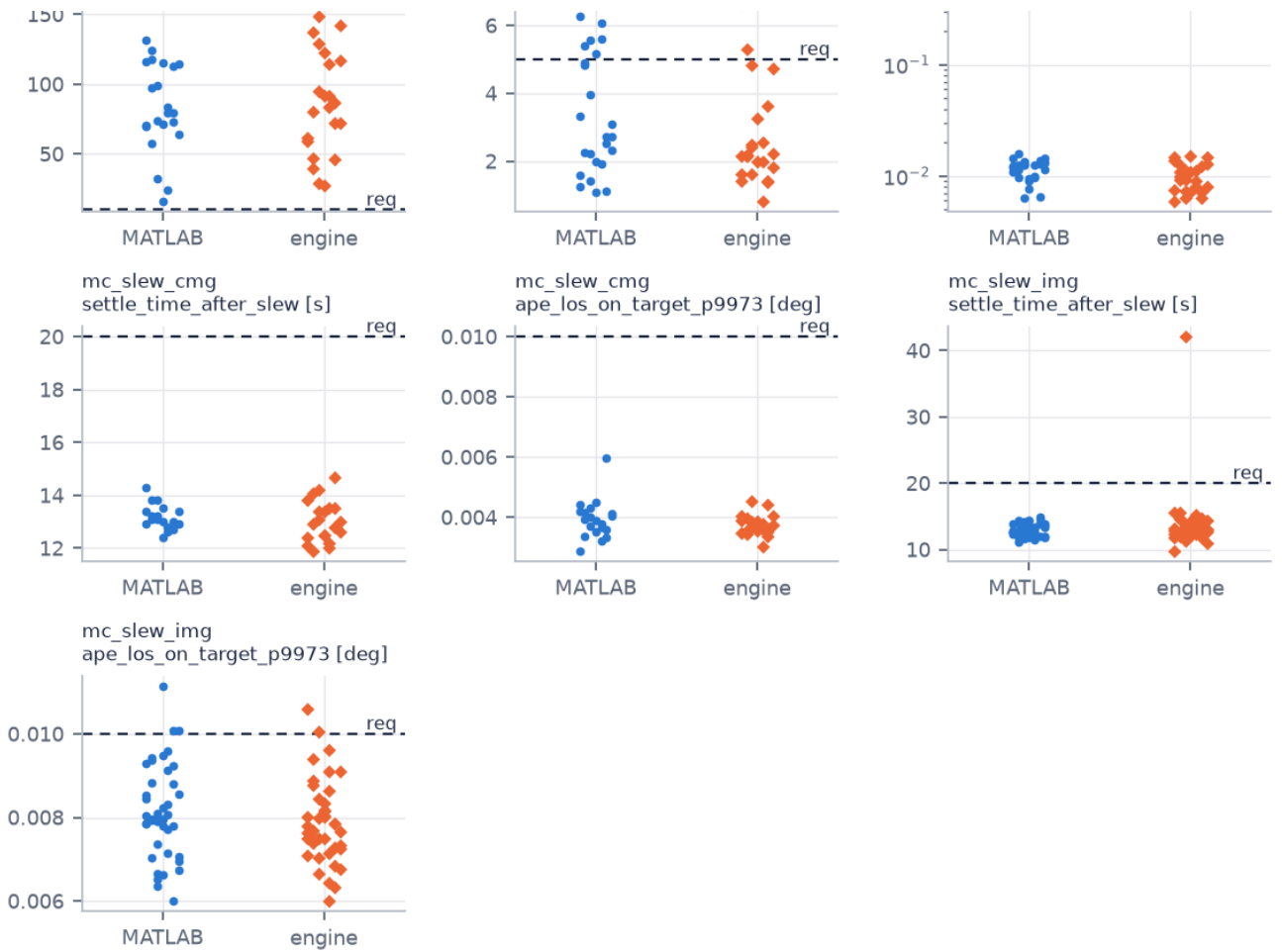


Figure 3. Every run of every campaign, requirement metrics: MATLAB twin (circles) and Rust engine (diamonds), dashed line = requirement.

9. Engine vs MATLAB twin parity

Single runs, engine vs MATLAB twin, same scenario, case and product: **94 of 107** requirement verdicts agree. The truth environment (orbit, Sun, Moon, shadow, density, field) of the engine's POP port is identical to the twin's on every filed sample of every scenario; what differs is the random stream (unit errors, initial tumble), so single-run metrics scatter inside the distributions of section 6. The differences traced to their cause are listed in results/ENGINE_PARITY.md.

scenario	samples	max r difference [m]	max density ratio – 1	max Sun-direction difference	max B
agile_slew_cmg	6001	0	0	0	
agile_slew_fmr_rcs	6001	0	0	0	
agile_slew_img	6001	0	0	0	
agile_slew_rw_rcs	6001	0	0	0	
agile_slew_vscmg	6001	0	0	0	
detumble_ais	17221	0	0	0	
detumble_ais_bangbang	17221	0	0	0	
detumble_ais_mag	17221	0	0	0	
detumble_img	17221	0	0	0	
fault_coil_ais	17221	0	0	0	
fault_gimbal_cmg	57401	0	0	0	
fault_gyro_ais	17221	0	0	0	
fault_st_img	57401	0	0	0	
fault_wheel_img	57401	0	0	0	
fine_hold_cmg	57401	0	0	0	
fine_hold_fmr	57401	0	0	0	
fine_hold_fmr_rcs	57401	0	0	0	
fine_hold_img	57401	0	0	0	
fine_hold_img_lqr	57401	0	0	0	
fine_hold_img_smc	57401	0	0	0	
fine_hold_rw_rcs	57401	0	0	0	
fine_hold_vscmg	57401	0	0	0	
mission_ais	28701	0	0	0	
mission_cmg	17221	0	0	0	
mission_fmr	17221	0	0	0	
mission_fmr_rcs	17221	0	0	0	
mission_img	17221	0	0	0	
mission_rw_rcs	17221	0	0	0	
mission_vscmg	17221	0	0	0	
nadir_hold_ais	17221	0	0	0	

scenario	samples	max r difference [m]	max density ratio – 1	max Sun-direction difference	max B
nadir_hold_ais_css	17221	0	0	0	
slew_cmgy	6001	0	0	0	
slew_fmry	6001	0	0	0	
slew_fmry_rcs	6001	0	0	0	
slew_img	6001	0	0	0	
slew_img_lqr	6001	0	0	0	
slew_img_smc	6001	0	0	0	
slew_rw_rcs	6001	0	0	0	
slew_vscmg	6001	0	0	0	
sun_spin_ais	17221	0	0	0	
agile_slew_cmgy	55	2.17	26x		
agile_slew_fmry_rcs	51	2.20	23x		
agile_slew_img	48	1.53	31x		
agile_slew_rw_rcs	55	1.52	36x		
agile_slew_vscmg	61	0.20	300x		
detumble_ais	496	0.97	511x		
detumble_ais_bangbang	848	0.74	1152x		
detumble_ais_mag	526	1.34	392x		
detumble_img	972	4.09	237x		
fault_coil_ais	533	0.59	908x		
fault_gimbal_cmgy	465	2.02	230x		
fault_gyro_ais	579	1.20	481x		
fault_st_img	389	1.74	223x		
fault_wheel_img	414	2.15	192x		
fine_hold_cmgy	468	2.58	182x		
fine_hold_fmry	659	1.77	371x		
fine_hold_fmry_rcs	523	2.28	230x		
fine_hold_img	390	1.38	282x		
fine_hold_img_lqr	388	1.91	203x		
fine_hold_img_smc	388	1.37	284x		
fine_hold_rw_rcs	394	1.72	229x		
fine_hold_vscmg	476	2.23	213x		
mission_ais	906	0.78	1158x		
mission_cmgy	1421	5.67	251x		
mission_fmry	1622	4.49	361x		
mission_fmry_rcs	1619	4.82	336x		
mission_img	1235	4.01	308x		
mission_rw_rcs	1263	4.95	255x		

scenario	samples	max r difference [m]	max density ratio – 1	max Sun-direction difference	max B
mission_vscmg	1461	5.92	247x		
nadir_hold_ais	569	0.64	892x		
nadir_hold_ais_css	550	0.66	836x		
slew_cmg	54	0.54	100x		
slew_fmr	79	0.22	358x		
slew_fmr_rcs	56	0.22	249x		
slew_img	46	0.62	74x		
slew_img_lqr	46	0.59	77x		
slew_img_smc	45	0.24	192x		
slew_rw_rcs	46	0.18	260x		
slew_vscmg	56	0.33	168x		
sun_spin_ais	501	0.62	809x		

Verdicts that differ (all traced; see ENGINE_PARITY.md):

scenario	metric	MATLAB	engine	req	MATLAB	engine
fine_hold_fmr	power_mean	4.348	0.08406	2	FAIL	pass
fine_hold_fmr_rcs	power_mean	4.367	0.08406	2	FAIL	pass
fine_hold_img	ape_los_p9973	0.009166	0.01049	0.01	pass	FAIL
fine_hold_rw_rcs	ape_los_p9973	0.008375	0.03388	0.01	pass	FAIL
fine_hold_rw_rcs	rate_stability_p9973	0.003604	0.005641	0.005	pass	FAIL
mission_img	ape_los_last_half_orbit_p9973	0.007627	0.01066	0.01	pass	FAIL
mission_vscmg	ape_los_last_half_orbit_p9973	0.01111	0.009595	0.01	FAIL	pass
nadir_hold_ais	ape_los_last_orbit	7.248	12.64	10	pass	FAIL
nadir_hold_ais	ake_los_last_orbit	3.864	5.491	5	pass	FAIL
slew_img_smc	settle_time_after_slew	3.2	—	20	pass	FAIL
sun_spin_ais	sun_spin_entry	40.44	142.4	95	pass	FAIL
sun_spin_ais	sun_angle_last_orbit	6.503	93.84	20	pass	FAIL
sun_spin_ais	spin_rate_error_last_orbit	0.04568	1.116	0.5	pass	FAIL

10. Soft OILS

Every scenario flown twice on the engine: SILS (C flight software in-process) and soft OILS (the flight software compiled for the OBC, arm-none-eabi-gcc -mcpu=cortex-m4 -mfloat-abi=hard, running as firmware on QEMU mps2-an386 behind adcs-link/1). In soft OILS each command reaches the actuators only after the OBC's sensor reads, its execution and its CAN frames; the execution is the step's exact instruction count (QEMU plugin) $\times$ CPI 1.25 / 168 MHz. **105 of 107** requirement verdicts agree between SILS and soft OILS over 40 scenarios; runs are deterministic.

scenario	period [ms]	instructions/step mean / max	exec max [ms]	latency mean / max [ms]	C
agile_slew_cmgy	100	3.5e+05 / 8.94e+05	6.65	4.2 / 8.24	
agile_slew_fmr_rcs	100	3.01e+05 / 8.45e+05	6.29	3.31 / 7.36	
agile_slew_img	100	2.94e+05 / 8.44e+05	6.28	3.14 / 7.22	
agile_slew_rw_rcs	100	3.04e+05 / 8.51e+05	6.33	3.34 / 7.4	
agile_slew_vscmg	100	3.68e+05 / 9.17e+05	6.82	4.33 / 8.41	
detumble_ais	200	1.19e+05 / 4.64e+05	3.45	1.44 / 4	
detumble_ais_bangbang	200	1.18e+05 / 4.64e+05	3.45	1.43 / 4	
detumble_ais_mag	200	1.19e+05 / 4.64e+05	3.45	1.44 / 4	
detumble_img	100	8.12e+04 / 4.89e+05	3.64	1.55 / 4.58	
fault_coil_ais	200	1.19e+05 / 4.64e+05	3.45	1.44 / 4	
fault_gimbal_cmgy	100	3.62e+05 / 8.98e+05	6.68	4.29 / 8.27	
fault_gyro_ais	200	1.93e+05 / 6.37e+05	4.74	1.99 / 5.3	
fault_st_img	100	2.54e+05 / 8.42e+05	6.26	2.84 / 7.21	
fault_wheel_img	100	2.9e+05 / 8.44e+05	6.28	3.1 / 7.22	
fine_hold_cmgy	100	3.62e+05 / 8.97e+05	6.68	4.29 / 8.27	
fine_hold_fmr	100	3.1e+05 / 8.45e+05	6.29	3.25 / 7.23	
fine_hold_fmr_rcs	100	3.13e+05 / 8.48e+05	6.31	3.4 / 7.39	
fine_hold_img	100	3.1e+05 / 8.45e+05	6.29	3.25 / 7.23	
fine_hold_img_lqr	100	3.1e+05 / 8.46e+05	6.29	3.25 / 7.24	
fine_hold_img_smc	100	3.13e+05 / 8.48e+05	6.31	3.27 / 7.26	
fine_hold_rw_rcs	100	3.15e+05 / 8.52e+05	6.34	3.42 / 7.42	
fine_hold_vscmg	100	3.83e+05 / 9.19e+05	6.84	4.44 / 8.43	
mission_ais	200	1.86e+05 / 7.58e+05	5.64	1.94 / 6.2	
mission_cmgy	100	3.05e+05 / 8.97e+05	6.67	3.87 / 8.27	
mission_fmr	100	2.61e+05 / 8.46e+05	6.3	2.89 / 7.24	
mission_fmr_rcs	100	2.64e+05 / 8.49e+05	6.32	3.04 / 7.39	
mission_img	100	2.68e+05 / 8.45e+05	6.29	2.94 / 7.23	
mission_rw_rcs	100	2.74e+05 / 8.55e+05	6.36	3.12 / 7.43	
mission_vscmg	100	3.23e+05 / 9.18e+05	6.83	3.99 / 8.42	

scenario	period [ms]	instructions/step mean / max	exec max [ms]	latency mean / max [ms]	C
nadir_hold_ais	200	1.93e+05 / 6.37e+05	4.74	1.99 / 5.3	
nadir_hold_ais_css	200	1.93e+05 / 6.38e+05	4.74	1.99 / 5.3	
slew_cmng	100	3.69e+05 / 8.94e+05	6.65	4.34 / 8.24	
slew_fmr	100	3.17e+05 / 8.41e+05	6.26	3.3 / 7.2	
slew_fmr_rcs	100	3.19e+05 / 8.44e+05	6.28	3.45 / 7.36	
slew_img	100	3.16e+05 / 8.42e+05	6.26	3.3 / 7.21	
slew_img_lqr	100	3.17e+05 / 8.42e+05	6.27	3.3 / 7.21	
slew_img_smc	100	3.19e+05 / 8.45e+05	6.29	3.32 / 7.23	
slew_rw_rcs	100	3.21e+05 / 8.47e+05	6.31	3.46 / 7.38	
slew_vscmg	100	3.89e+05 / 9.15e+05	6.81	4.49 / 8.4	
sun_spin_ais	200	1.21e+05 / 4.65e+05	3.46	1.45 / 4.01	

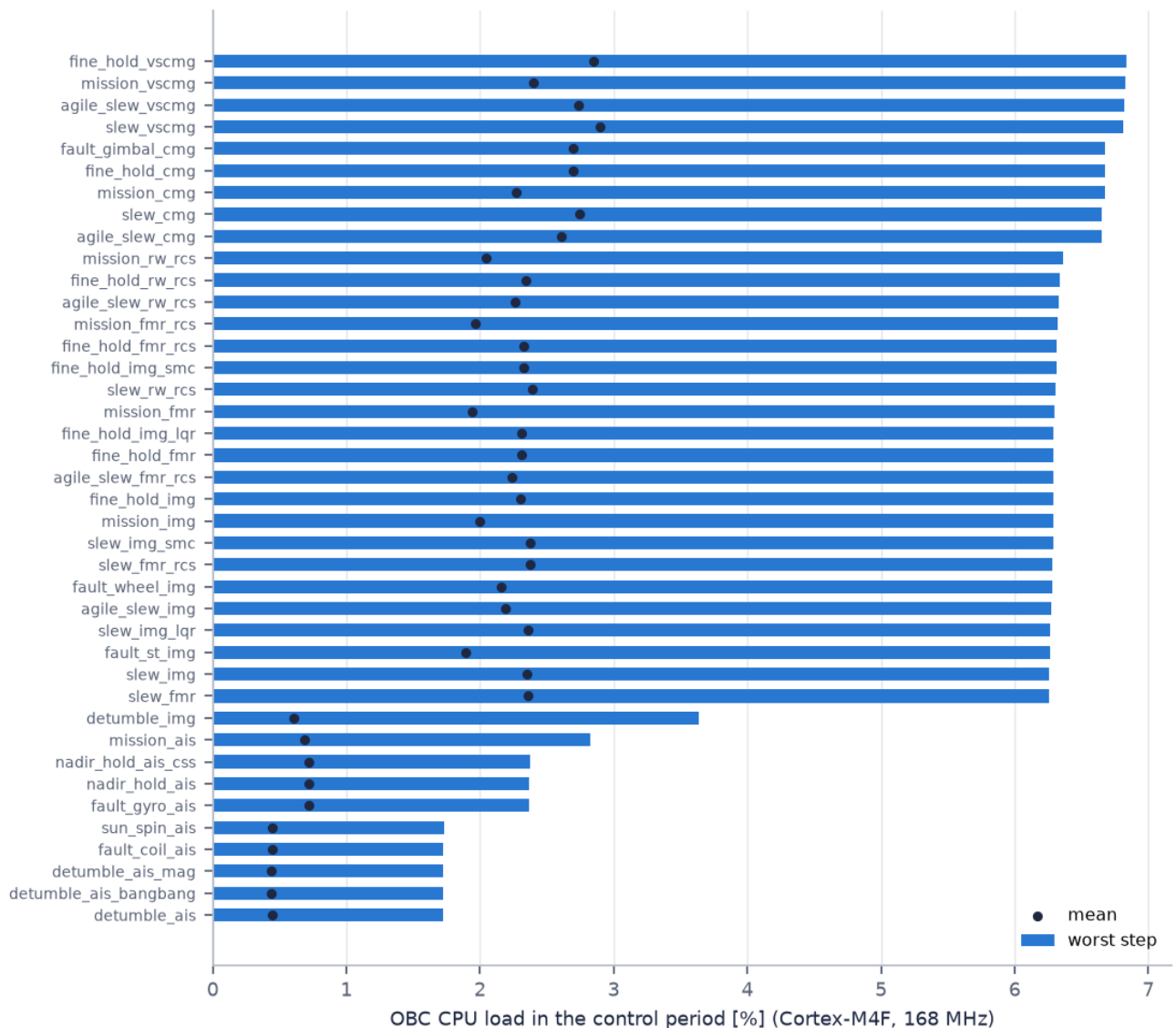


Figure 4. OBC load per scenario: worst step (bar) and mean (dot).

Verdicts that change between SILS and soft OILS (the command latency's effect):

scenario	metric	req	SILS	soft OILS	SILS	soft OILS
agile_slew_vscmg	ape_los_on_target_p9973	0.01	0.01008	0.009977	FAIL	pass
sun_spin_ais	sun_spin_entry	95	142.4	59.66	FAIL	pass

11. Virtual OBC bit-identity

Zero-latency lockstep (no timing model): the flight software as a host process and as QEMU Cortex-M4F firmware, C and Rust, reproduces the in-process trajectory bit for bit: **35 of 35** runs bit-identical.

scenario	reference	virtual OBC	result	wall [s]
detumble_ais	c (in-process)	obc-posix	bit-identical	0.3
detumble_ais	rust (in-process)	obc-posix-rs	bit-identical	0.2
detumble_ais	c (in-process)	qemu	bit-identical	18.2
detumble_ais	rust (in-process)	qemu-rs	bit-identical	17.4
detumble_ais	c (in-process)	rust	bit-identical	0.2
mission_ais	c (in-process)	obc-posix	bit-identical	0.2
mission_ais	rust (in-process)	obc-posix-rs	bit-identical	0.2
mission_ais	c (in-process)	qemu	bit-identical	14.2
mission_ais	rust (in-process)	qemu-rs	bit-identical	15.4
mission_ais	c (in-process)	rust	bit-identical	0.1
fine_hold_img	c (in-process)	obc-posix	bit-identical	0.6
fine_hold_img	rust (in-process)	obc-posix-rs	bit-identical	0.5
fine_hold_img	c (in-process)	qemu	bit-identical	55.9
fine_hold_img	rust (in-process)	qemu-rs	bit-identical	61.1
fine_hold_img	c (in-process)	rust	bit-identical	0.3
slew_img	c (in-process)	obc-posix	bit-identical	0.6
slew_img	rust (in-process)	obc-posix-rs	bit-identical	0.5
slew_img	c (in-process)	qemu	bit-identical	62.8
slew_img	rust (in-process)	qemu-rs	bit-identical	63.3
slew_img	c (in-process)	rust	bit-identical	0.3
fine_hold_cmng	c (in-process)	obc-posix	bit-identical	0.6
fine_hold_cmng	rust (in-process)	obc-posix-rs	bit-identical	0.6
fine_hold_cmng	c (in-process)	qemu	bit-identical	64.2
fine_hold_cmng	rust (in-process)	qemu-rs	bit-identical	68.9
fine_hold_cmng	c (in-process)	rust	bit-identical	0.3
fine_hold_fmr_rcs	c (in-process)	obc-posix	bit-identical	0.4
fine_hold_fmr_rcs	rust (in-process)	obc-posix-rs	bit-identical	0.4
fine_hold_fmr_rcs	c (in-process)	qemu	bit-identical	57.5
fine_hold_fmr_rcs	rust (in-process)	qemu-rs	bit-identical	57.1
fine_hold_fmr_rcs	c (in-process)	rust	bit-identical	0.5

scenario	reference	virtual OBC	result	wall [s]
sun_spin_ais	c (in-process)	obc-posix	bit-identical	0.2
sun_spin_ais	rust (in-process)	obc-posix-rs	bit-identical	0.2
sun_spin_ais	c (in-process)	qemu	bit-identical	15.3
sun_spin_ais	rust (in-process)	qemu-rs	bit-identical	13.6
sun_spin_ais	c (in-process)	rust	bit-identical	0.1

12. Open items

- Bang-bang B-dot (fixed): pure sign switching limit-cycled around the 0.5 deg/s exit rate in 4 of 12 seeds with or without OBC latency; a boundary layer (4 x the B-dot gain inside it) detumbles 12 of 12 at 0, 4 and 8 ms. C, Rust, MATLAB and pseudocode updated.
- Rate stability (imaging): relaxed by the owner from 0.001 to 0.005 deg/s (18 arcsec/s, ~0.5 m smear in a 10 ms exposure at 550 km). At 0.001 deg/s it was set by gyro noise for every actuator (0.0025-0.004 deg/s with TRN-GYRO-P1) and needed a fibre-optic-class gyro (0.6 kg); at 0.005 deg/s the catalogue gyro holds it. The ADCS budget is 1.6 kg / 1.0 L (1U of the 3U; ceiling 1.7 kg).
- Fluid loop fine pointing: limited by the loop's flow sensor, not the pump. The converged imaging design carries a 0.125 mm/s (1 sigma) flow sensor; this is a requirement on the in-house sensor. One star-tracker head instead of two breaks nadir pointing, so both stay.
- Engine vs MATLAB on fine_hold_rw_rcs rate stability: 0.0056 against 0.0036 deg/s (ratio 1.56, as before the relaxation); the new 0.005 deg/s line now falls between them, so this verdict disagrees. The gap is the wheel-plus-RCS model, not the requirement.
- Sun spin: 17 of 24 seeds pass with and without the OBC; failures are Sun-spin entry with the Sun near the XY plane tripping the 1 deg/s exit guard during the L2 precession transient -- a tuning trade (fsw.sun_spin_perp_out_dps, fsw.sun_spin_dwell_out_s).
- Bought against designed (both cases, same budget of 1.6 kg and 1.0 L): on ais_img_3u three CubeSpace CW0017 wheels are feasible at 1.0 kg against our fluid loop's 1.6 kg, so the lightest configuration overall is a benchmark; the fluid loop is selected because the selection is among our solutions, and its 0.6 kg is the price of not buying wheels. On ais_3u every rotor fails the 0.5 W orbit-average power (three CW0017 draw 0.9 W steady) and the fluid loop, feasible at 1.45 kg, is the only one that passes. The Tensor Tech CMG cluster fails power in both cases on its whole-ADCS datasheet figure (4 W upper bound).
- Catalogue data: the vendor sites were not reachable from the build environment. CubeSpace Gen2, Rocket Lab RW-0.01, RW3-0.06, RW3-1.0 and Tensor Tech ADCS100/400 numbers were read from the full datasheets (distributor mirror); AAC Clyde Space RW222/RW400, Rocket Lab RW-0.03 and Tensor Tech CMG-10m from search excerpts. AAC Clyde Space models lack mass or power on what could be read, so they are listed but not selectable; confirm with the vendors before a buy decision.
- Soft OILS models the OBC's CPU and buses; real OILS still needs the board target (fsw/targets/<board>/main.c: UART/Ethernet, clock, linker script, modelled on fsw/targets/qemu-mps2) and --realtime on adcs-link/1.